

## Scalability & Future Plan

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|---------------|-------------------------------------------------------------|
| Date          | 15 July 2026                                                |
| Team ID       | XXXXXX                                                      |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine |
| Maximum Marks | 1 Mark                                                      |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation                                                                  | Impact                                                                      | Priority to Address<br>(High / Medium / Low) |
|------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------|----------------------------------------------|
| 1    | Recommendations are limited to the crops available in the training dataset. | Cannot recommend new or region-specific crops without retraining the model. | High                                         |
| 2    | Weather information must be entered manually by the user.                   | Incorrect manual inputs may reduce prediction accuracy.                     | High                                         |
| 3    | No user authentication or cloud database integration.                       | User data and prediction history cannot be stored securely.                 | Medium                                       |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                                                  | Proposed Upgrade / Solution                                                                                                                        |
|------|--------------------|--------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports a limited number of local users through the Flask development server. | Deploy on a cloud platform (AWS, Azure, Render, or Railway) using a production-ready WSGI server such as Gunicorn or Waitress with load balancing. |
| 2    | Data Storage       | Uses a static dataset and a Pickle model file without persistent storage.      | Integrate a relational database such as MySQL or PostgreSQL to store user information, prediction history, and updated datasets.                   |
| 3    | Performance        | Predictions are generated using a locally loaded machine learning model.       | Optimize preprocessing, cache the model, and use optimized inference techniques for faster response times.                                         |
| 4    | Security           | No authentication or encrypted communication.                                  | Implement user login, HTTPS, input validation, and role-based access control to improve application security.                                      |

Future Roadmap:

| Phase   | Planned Feature / Enhancement                 | Target Timeline | Expected Impact                                                                                           |
|---------|-----------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------------|
| Phase 1 | Fertilizer Recommendation System              | 1–2 Months      | Help farmers choose suitable fertilizers along with crop recommendations.                                 |
| Phase 2 | Weather API Integration                       | 2–3 Months      | Automatically fetch real-time weather data and improve prediction accuracy.                               |
| Phase 3 | Crop Disease Detection using Image Processing | 3–5 Months      | Enable early disease detection and provide preventive measures to farmers.                                |
| Phase 4 | Mobile Application with Multilingual Support  | 5–6 Months      | Improve accessibility for farmers by providing a user-friendly mobile app in multiple regional languages. |